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Clock-Gating and Its Application to Low Power Design of Sequential Circuits

Qing Wu, Massoud Pedram, and Xunwei Wu

***Abstract*—This paper models the clock behavior in a sequential circuit by a quaternary variable and uses this representation to propose and analyze two clock-gating techniques. It then uses the covering relationship between the triggering transition of the clock and the active cycles of various flip flops to generate a derived clock for each flip flop in the circuit. A technique for clock gating is also presented, which generates a derived clock synchronous with the master clock. Design examples using gated clocks are provided next. Experimental results show that these designs have ideal logic functionality with lower power dissipation compared to traditional designs.**

***Index Terms*—Clock gating, CMOS, logic, low power, sequential circuit, synthesis.**

I. Introduction

The sequential circuits in a system are considered major contributors to the power dissipation since one input of sequential circuits is the clock, which is the only signal that switches all the time. In addition, the clock signal tends to be highly loaded. To distribute the clock and control the clock skew, one needs to construct a clock network (often a clock tree) with clock buffers. All of this adds to the capacitance of the clock net. Recent studies indicate that the clock signals in digital computers consume a large (15–45%) percentage of the system power [1]. Thus, the circuit power can be greatly reduced by reducing the clock power dissipation.

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Most efforts for clock power reduction have focused on issues such as reduced voltage swings, buffer insertion, and clock routing [2]. In many cases, switching of the clock causes a great deal of unnecessary gate activity. For that reason, circuits are being developed with controllable clocks. This means that from the master clock other clocks are derived which, based on certain conditions, can be slowed down or stopped completely with respect to the master clock. Obviously, this scheme results in power savings due to the following factors.

1) The load on the master clock is reduced and the number of required buffers in the clock tree is decreased. Therefore, the power dissipation of clock tree can be reduced.
2) The flip flop receiving the derived clock is not triggered in idle cycles and the corresponding dynamic power dissipation is thus saved.
3) The excitation function of the flip flop triggered by the derived clock may be simplified since it has a do not care condition in the cycle when the flip flop is not triggered by the derived clock.

The clock-gating problem has been studied in [3]–[5]. In [3] the authors presented a technique for saving power in the clock tree by stopping the clock fed into idle modules. However, a number of engineering issues related to the design of the clock tree were not addressed and, hence, the proposed approach has not been adopted in practice. In [4], a precomputation-based technique is used to generate a signal to control the load enable pin of the flip flops in the data path. The control signal is derived by investigating the relationship between the latched input and the primary outputs of the combinational blocks in the data path. The technique is useful only if the outputs of the block can be precomputed (predicted) for certain input assignments. In [5], the authors use a latch to gate the clock in control-dominated circuits. The problem is that the additional latch receives the clock's triggering signal, which results in extra power dissipation in the latch itself. Besides, this scheme results in the derived clock having a considerable skew with respect to the master clock.

This paper investigates various issues in deriving a gated clock from a master clock. In Section II, a quaternary variable is used to model the clock behavior and to discuss its triggering action on flip flops. Based on this analysis, two clock-gating schemes are proposed. In Section III, we use the covering relation between the clock and the transition behaviors of the triggered flip flops to derive conditions for gating the master clock. Two common sequential circuits, i.e., 8421 BCD code up-counter and three-excess counter, are then described to illustrate the procedure for finding a derived clock. In Section IV, a new technique for clock gating is presented which generates a clock synchronous with the master clock. This eliminates the additional skew between the master clock and the derived clock. Thus, the designed sequential circuit is a synchronous one. Finally, we present circuit simulation results to prove the quality of the derived clock and its ability to reduce power dissipation in the circuit.

II. Description for Clock Behavior and Clock-Gating

In a synchronous system, a flip flop is triggered by a certain directional transition of a clock signal. For the clock to be another signal rather than the master clock, it must offer the same directional transition to trigger the flip flop and it must be in step with the master clock.

For the clock signal clk in a circuit if we denote its logic values before and after a transition as $clk(t)$ and $clk^+(t)$, respectively, four combinations can be used to express different behaviors of the clock as shown in Table I, where a special quaternary variable $\tilde{clk}$ denotes the corresponding behavior. The four values are $(0,\ \alpha,\ \beta,\ 1)$, where $\alpha,\ \beta$ represent two kinds of transition behaviors and 0, 1 represent two kinds

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of holding behaviors. (Note that although they have the same forms as signal values 0 and 1, their meanings are different.)

In addition, we can also define a literal operation to identify the behavior of a clock

$$clk^b = \begin{cases} 1, & \text{if } \tilde{clk} = b \\ 0, & \text{if } \tilde{clk} \neq b \end{cases} \tag{1}$$

where $b \in \{0, \alpha, \beta, 1\}$. Thus, the rising transition clk^α and the falling transition clk^β of a clock are binary variables and can serve as arguments of Boolean operations. For example, from Table I we have

$$clk^0 = \overline{clk} \cdot \overline{clk^+} \quad clk^\alpha = \overline{clk} \cdot clk^+$$
$$clk^\beta = clk \cdot \overline{clk^+} \quad \text{and} \quad clk^1 = clk \cdot clk^+.$$

Assume that there are n flip flops in a sequential circuit and that their outputs and clock inputs are denoted by Q_i and clk_i, $i = 0, 1, \cdots, n-1$, respectively. For a synchronous sequential circuit we have $clk_i = clk$, namely, all flip flops are triggered by the same master clock signal clk. However, if a flip flop Q_i is to be disconnected from the master clock during some (idle) cycles, then we have to use a derived clock for Q_i. Notice that this derived clock should be in step with the master clock for the circuits to remain synchronous.

Generally, we consider that the derived clock is obtained from the master clock clk and the outputs of other flip flops $Q_0, \cdots, Q_{i-1}, Q_{i+1}, \cdots, Q_{n-1}$ (which make transitions following the triggering transition of their respective clocks.) Since both AND gating and OR gating can be used for controlling the master clock, we have the following two clock-gating forms:

$$clk_i = g_i + p_i \cdot clk \tag{2}$$

$$clk_i = g_i \cdot (p_i + clk) \tag{3}$$

where g_i and p_i are functions of flip-flop outputs $Q_0, \cdots, Q_{i-1}, Q_{i+1}, \cdots, Q_{n-1}$. Furthermore, we assume that the delay on the p_i's and g_i's is shorter than the clock pulse width. Consider a flip flop triggered by the falling clock transition as an example (i.e., a negative edge-triggered flip flop). The timing relationships of clk, p_i, $p_i \cdot clk$, and $p_i + clk$ are shown in Fig. 1. Note that p_i which exhibits a delay with respect to the falling transition of the clock, may have glitches (represented by vertical grid lines) and has its final stable value in the zone where $clk = 0$. We can see that $p_i + clk$ cannot prevent the glitches and may even lead to an extra glitch. Therefore, only (2) is suitable for the negative edge-triggered flip flops while (3) is not. Note that g_i in (2) must be glitch-free when $clk = 0$. The above discussion shows that the falling transition of clk_i in (2) occurs for the following two cases.

1) When $g_i = 0$ and $p_i = 1$, falling transition of clk leads to falling transition of the derived clock clk_i. Therefore, p_i may be named the transition propagate term.
2) When $g_i = 1$ and g_i make a falling transition, the derived clock clk_i makes a falling transition since clk and, hence, $p_i \cdot clk$ are 0 at that time instance. Therefore, g_i may be named the transition generate term.

From this analysis, we obtain

$$clk_i^\beta = g_i^\beta + \overline{g_i} \cdot p_i \cdot clk^\beta. \tag{4}$$

Similarly, we can show that the derived clock signal in (3) is suitable for the flip flops triggered by the rising transition of the clock. Here, g_i in (3) must be glitch free when $clk = 1$. The rising transition of clk_i can be expressed as

$$clk_i^\alpha = g_i^\alpha + g_i \cdot \overline{p_i} \cdot clk^\alpha. \tag{5}$$

TABLE I
QUATERNARY REPRESENTATION FOR BEHAVIORS OF A SIGNAL

$\tilde{clk}$	$clk(t) \rightarrow clk^+(t)$		Behavior
0	0	0	0-holding
α	0	1	α-transition
β	1	0	β-transition
1	1	1	1-holding

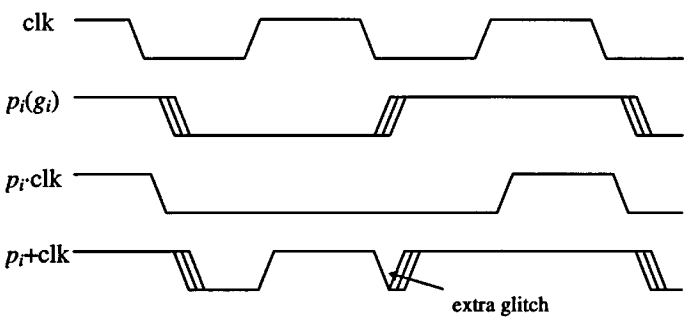


Fig. 1. Timing relationship of clk, $p_i(g_i)$, $p_i \cdot clk$ and $p_i + clk$.

TABLE II
NEXT STATES AND STATE BEHAVIORS OF AN 8421 BCD CODE UP COUNTER

Q_3	Q_2	Q_1	Q_0	Q_3^+	Q_2^+	Q_1^+	Q_0^+	$\tilde{Q}_3$	$\tilde{Q}_2$	$\tilde{Q}_1$	$\tilde{Q}_0$
0	0	0	0	0	0	0	1	0	0	0	α
0	0	0	1	0	0	1	0	0	0	α	β
0	0	1	0	0	0	1	1	0	0	1	α
0	0	1	1	0	1	0	0	0	α	β	β
0	1	0	0	0	1	0	1	0	1	0	α
0	1	0	1	0	1	1	0	0	1	α	β
0	1	1	0	0	1	1	1	0	1	1	α
0	1	1	1	1	0	0	0	α	β	β	β
1	0	0	0	1	0	0	1	1	0	0	α
1	0	0	1	0	0	0	0	β	0	0	β

It should be pointed out that the attached circuitry needed for generating the derived clock should be simple to avoid excessive power dissipation due to this overhead circuitry. Therefore, g_i and p_i in (2) and (3) should be relatively simple functions. Especially, we require g_i to be simple to avoid dangerous glitches. Note that if $g_i = 0$, $p_i = 1$ in (2) or $g_i = 1$, $p_i = 0$ in (3), we return to the condition of applying the master clock clk in a synchronous sequential circuit.

III. DESIGN OF SEQUENTIAL CIRCUITS BASED ON DERIVED CLOCK

Assume that the derived clock for the flip flop Q_i is clk_i. Falling transitions of clk_i have to cover all cycles when the flip flop Q_i makes transitions Q_i^α and Q_i^β. The covering relation can be expressed as

$$clk_i^\beta \geq (Q_i^\alpha + Q_i^\beta). \tag{6}$$

Since AND and OR operations on Boolean variables can be interpreted as minimum and maximum operations on these variables, i.e., $x \cdot y = \min(x, y)$ and $x + y = \max(x, y)$, we can obtain the following equations from (6):

$$clk_i^\beta \cdot (Q_i^\alpha + Q_i^\beta) = (Q_i^\alpha + Q_i^\beta) \tag{7}$$

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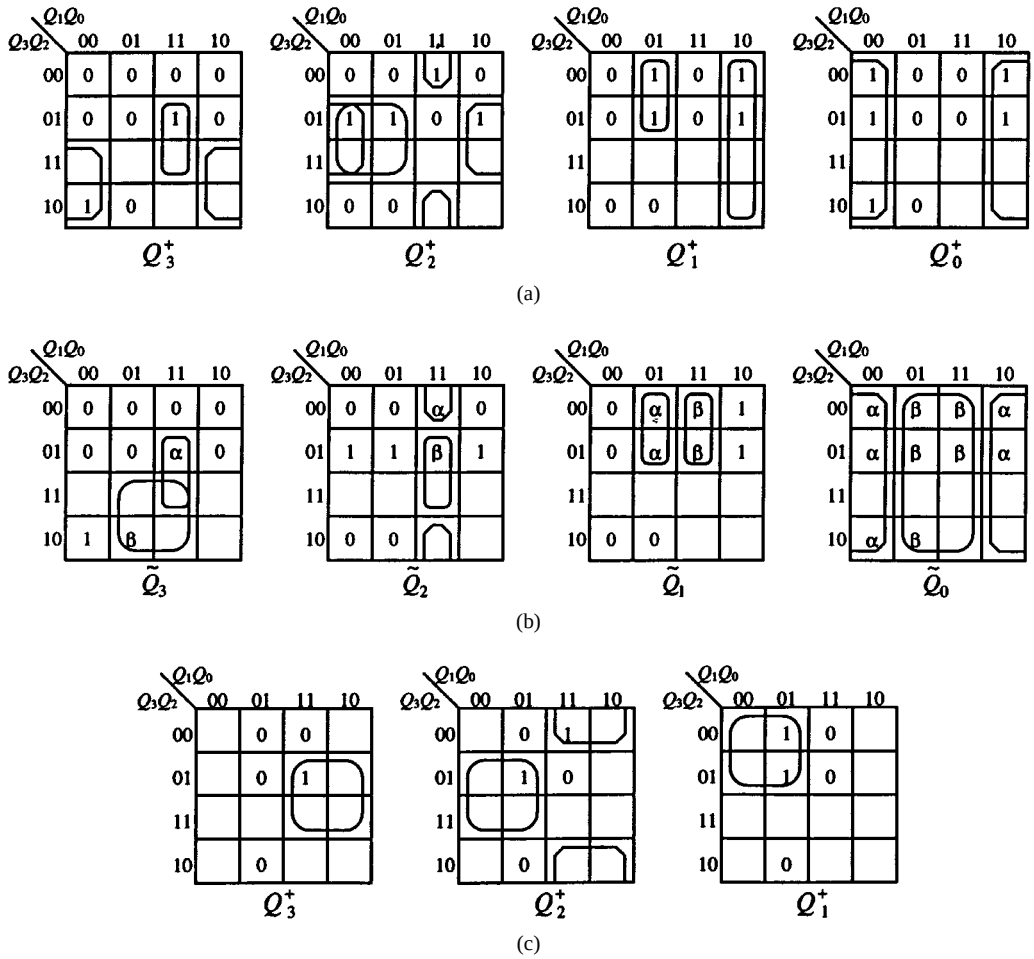


Fig. 2. (a) Next state Karnaugh maps, (b) behavior Karnaugh maps, and (c) simplified next state Karnaugh maps.

$$clk_i^{\beta} + (Q_i^{\alpha} + Q_i^{\beta}) = clk_i^{\beta}. \tag{8}$$

Therefore, we should first obtain $(Q_i^{\alpha} + Q_i^{\beta})$ and then generate the derived clock clk_i for flip flop Q_i. We will show the procedure by using design examples.

Example 1—Design of an 8421 BCD Code Up Counter: The next states and state behaviors of an 8421 BCD code up counter are shown in Table II, where behavior of each flip flop ($Q_i \rightarrow Q_i^+$) is denoted by $\tilde{Q}_i$. From Table II, the corresponding next state Karnaugh maps and behavior Karnaugh maps may be obtained, as shown in Fig. 2(a) and (b). In these maps, an empty box represents the don't care condition. The two transition functions for each flip flop can be derived from their corresponding behavior Karnaugh maps as below

$$Q_3^{\alpha} = Q_2 \cdot Q_1 \cdot Q_0 \cdot clk^{\beta}, \quad Q_3^{\beta} = Q_3 \cdot Q_0 \cdot clk^{\beta} \tag{9}$$

$$Q_2^{\alpha} = \overline{Q_2} \cdot Q_1 \cdot Q_0 \cdot clk^{\beta}, \quad Q_2^{\beta} = Q_2 \cdot Q_1 \cdot Q_0 \cdot clk^{\beta} \tag{10}$$

$$Q_1^{\alpha} = \overline{Q_3} \cdot \overline{Q_1} \cdot Q_0 \cdot clk^{\beta}, \quad Q_1^{\beta} = \overline{Q_3} \cdot Q_1 \cdot Q_0 \cdot clk^{\beta} \tag{11}$$

$$Q_0^{\alpha} = \overline{Q_0} \cdot clk^{\beta}, \quad Q_0^{\beta} = Q_0 \cdot clk^{\beta}. \tag{12}$$

Therefore, we have

$$Q_3^{\alpha} + Q_3^{\beta} = (Q_3 + Q_2 \cdot Q_1) \cdot Q_0 \cdot clk^{\beta} \tag{13}$$

$$Q_2^{\alpha} + Q_2^{\beta} = Q_1 \cdot Q_0 \cdot clk^{\beta} \tag{14}$$

$$Q_1^{\alpha} + Q_1^{\beta} = \overline{Q_3} \cdot Q_0 \cdot clk^{\beta} \tag{15}$$

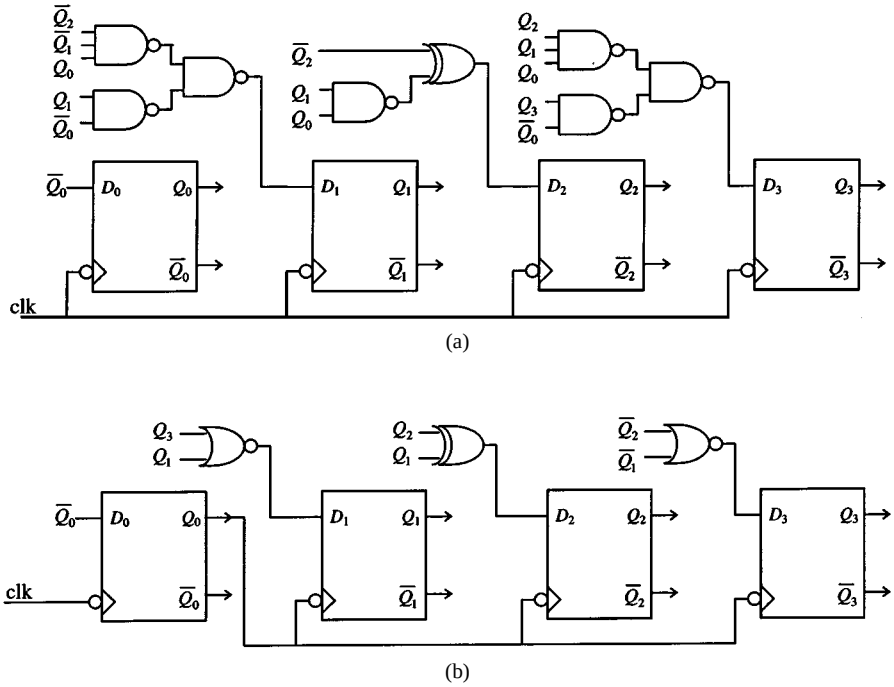
$$Q_0^{\alpha} + Q_0^{\beta} = clk^{\beta}. \tag{16}$$

From (13)–(15) we find that $Q_0 \cdot clk^{\beta} \geq (Q_i^{\alpha} + Q_i^{\beta})$, $(i = 1, 2, 3)$. From (12), we see that $Q_0^{\beta} = Q_0 \cdot clk^{\beta}$ can serve as the needed falling transition trigger for flip flops Q_1, Q_2, and Q_3, namely $clk_1^{\beta} = clk_2^{\beta} = clk_3^{\beta} = Q_0^{\beta}$. Comparing these with (4), we get $g_i = Q_0$, $p_i = 0$, and $clk_i = Q_0$ $(i = 1, 2, 3)$.

As for clk_0, (16) indicates that the clock for Q_0 is no other than the master clock clk. Since we only need to take care of the excitation input when the flip flop receives a triggering falling clock transition (i.e., β entries in $\tilde{Q}_0$ map), we do not care what the excitation inputs in other conditions are. Therefore, the next state Karnaugh maps for flip flops Q_1, Q_2, and Q_3 in Fig. 2(a) can be simplified to those shown in Fig. 2(c).

From Fig. 2(a) and (c) we can get the corresponding both synchronous and asynchronous designs, as shown in Fig. 3. (We say asynchronous, because now not all flip flops are triggered at the same time.) Obviously, the corresponding combinational circuits are simpler. Furthermore, since three flip flops Q_3, Q_2, Q_1 have no

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(a)

(b)

Fig. 3. Circuit realizations of BCD code up counter: (a) Synchronous design. (b) Asynchronous design.

TABLE III
THE NEXT STATES AND STATE BEHAVIORS OF AN EXCESS-THREE CODE UP COUNTER

Q_3	Q_2	Q_1	Q_0	Q_3^+	Q_2^+	Q_1^+	Q_0^+	$\tilde{Q}_3$	$\tilde{Q}_2$	$\tilde{Q}_1$	$\tilde{Q}_0$
0	0	1	1	0	1	0	0	0	α	β	β
0	1	0	0	0	1	0	1	0	1	0	α
0	1	0	1	0	1	1	0	0	1	α	β
0	1	1	0	0	1	1	1	0	1	1	α
0	1	1	1	1	0	0	0	α	β	β	β
1	0	0	0	1	0	0	1	1	0	0	α
1	0	0	1	1	0	1	0	1	0	α	β
1	0	1	0	1	0	1	1	1	0	1	α
1	0	1	1	1	1	0	0	1	α	β	β
1	1	0	0	0	0	1	1	β	β	α	α

dynamic power dissipation half of the time when there is no clock triggering, and because the simpler combinational circuits has lower node capacitance, the asynchronous design is saving power.

Example 2—Design of an Excess-Three Code Up Counter: The next state and state transition of an excess-three code up counter are shown in Table III. Transition functions for each flip flop can be derived as below

$$Q_3^\alpha = Q_2 \cdot Q_1 \cdot Q_0 \cdot clk^\beta, \quad Q_3^\beta = Q_3 \cdot Q_2 \cdot \overline{Q_1} \cdot clk^\beta \tag{17}$$

$$Q_2^\alpha = \overline{Q_2} \cdot Q_1 \cdot Q_0 \cdot clk^\beta,$$
$$Q_2^\beta = (Q_3 \cdot Q_2 \cdot \overline{Q_1} + Q_2 \cdot Q_1 \cdot Q_0) \cdot clk^\beta \tag{18}$$

$$Q_1^\alpha = (Q_3 \cdot Q_2 \cdot \overline{Q_0} + \overline{Q_1} \cdot Q_0) \cdot clk^\beta, \quad Q_1^\beta = Q_1 \cdot Q_0 \cdot clk^\beta \tag{19}$$

$$Q_0^\alpha = \overline{Q_0} \cdot clk^\beta, \quad Q_0^\beta = Q_0 \cdot clk^\beta. \tag{20}$$

Therefore, we have

$$Q_3^\alpha + Q_3^\beta = (Q_3 \cdot Q_2 \cdot \overline{Q_1} + Q_2 \cdot Q_1 \cdot Q_0) \cdot clk^\beta = Q_2^\beta \tag{21}$$

$$\begin{aligned} Q_2^\alpha + Q_2^\beta &= (Q_3 \cdot Q_2 \cdot \overline{Q_1} + Q_1 \cdot Q_0) \cdot clk^\beta \\ &= Q_1^\beta + \overline{Q_1} \cdot (Q_3 \cdot Q_2) clk^\beta \end{aligned} \tag{22}$$

$$Q_1^\alpha + Q_1^\beta = (Q_3 \cdot Q_2 \cdot \overline{Q_0} + Q_0) \cdot clk^\beta = Q_0^\beta + \overline{Q_0} \cdot (Q_3 \cdot Q_2) \cdot clk^\beta \tag{23}$$

$$Q_0^\alpha + Q_0^\beta = clk^\beta. \tag{24}$$

Based on (2) and (4), (22) and (23) can be reexpressed as

$$Q_2^\alpha + Q_2^\beta = [Q_1 + (Q_3 \cdot Q_2) \cdot clk]^\beta \tag{25}$$

$$Q_1^\alpha + Q_1^\beta = [Q_0 + (Q_3 \cdot Q_2) \cdot clk]^\beta. \tag{26}$$

Obviously, if we take $clk_3 = Q_2$, $clk_2 = [Q_1 + (Q_3 \cdot Q_2) \cdot clk]$, $clk_1 = [Q_0 + (Q_3 \cdot Q_2) \cdot clk]$, and $clk_0 = clk$, the covering relation will set the excitation functions of all the four flip flops as $D_i = \overline{Q_i}$ $(i = 0, 1, 2, 3)$. On the other hand, if we use the master clock for triggering all four flip flops, we obtain the following complicated excitation functions:

$$\begin{aligned} D_3 &= Q_2 \cdot Q_1 \cdot Q_0 + Q_3 \cdot \overline{Q_2}, \\ D_2 &= \overline{Q_2} \cdot Q_1 \cdot Q_0 + \overline{Q_3} \cdot \overline{Q_1} + \overline{Q_3} \cdot \overline{Q_0}, \\ D_1 &= Q_3 \cdot Q_2 + \overline{Q_1} \cdot Q_0 + Q_1 \cdot \overline{Q_0}, \\ D_0 &= \overline{Q_0}. \end{aligned}$$

Since the above D_3, D_2, and D_1 have complicated forms, their corresponding synchronous circuit realization will have a complicated combinational circuit with more node capacitance and, hence, higher power dissipation. On the other hand, the corresponding asynchronous circuit realization with $D_i = \overline{Q_i}$ is much simpler. There is power saving since

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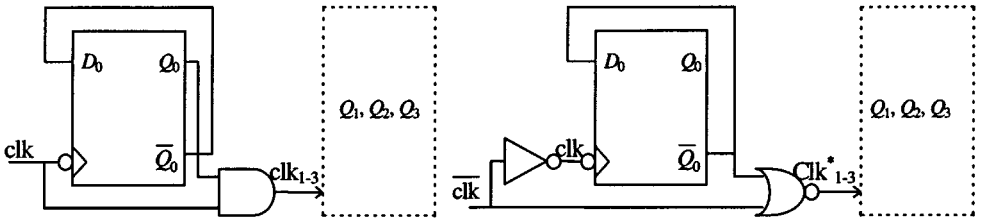


Fig. 4. BCD code up- counter by gating clock. (a) Asynchronous design. (b) Synchronous design.

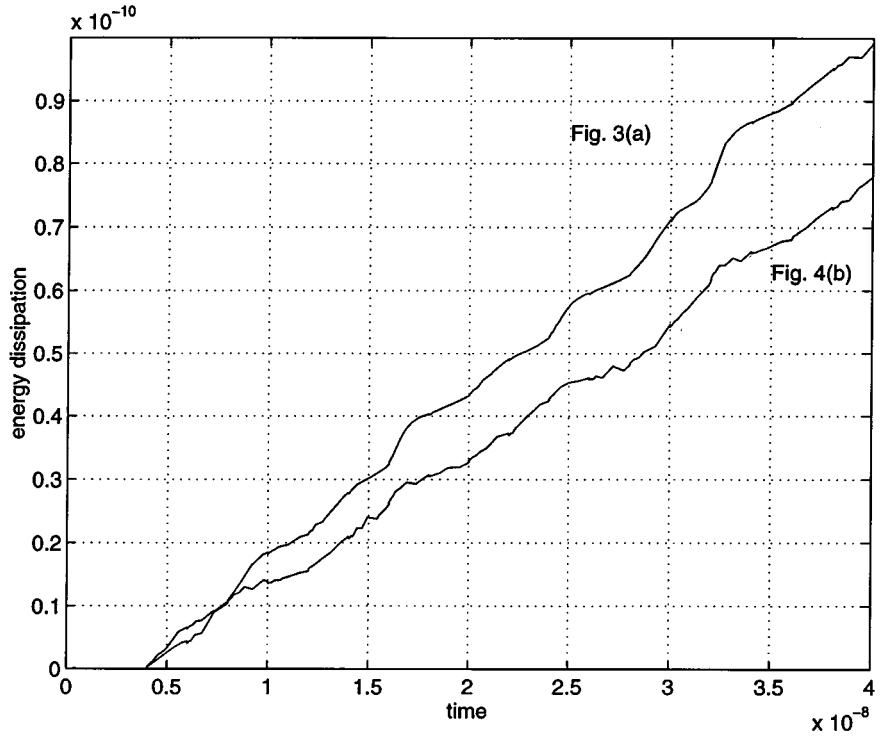


Fig. 5. Power dissipation diagram.

the four flip flops are isolated from the triggering clock in the idle cycles.

IV. Synchronous Derived Clock and Its Application

In Example 1 of Section III, we take $clk_i^\beta = Q_0^\beta, (i = 1, 2, 3)$. From (12) we can also write clk_i^β as $clk_i^\beta = Q_0 \cdot clk^\beta, (i = 1, 2, 3)$. Comparing this with (4), we have $g_i = 0, p_i = Q_0$, and $clk_i = Q_0 \cdot clk$. According to this form of the derived clock we get another asynchronous design, as shown in Fig. 4(a). At first glance, the circuit has one AND gate more than the design in Fig. 3(b). Furthermore, it appears that the derived clock $clk_{1\text{-}3}$ may have an increased phase delay. However, the timing relation shown in Fig. 1 indicates that the transition delay of $clk_{1\text{-}3}$ is independent of the delay of the Q_0 output. The delay between clk and $clk_{1\text{-}3}$ is only $2t_g$ (t_g is the average delay of a gate), which is less than the delay of the flip-flop output. Based on the above discussion, we can rewrite $clk_i = Q_0 \cdot clk$ as $clk_i^* = \overline{\overline{Q_0} + \overline{clk}}$. Furthermore, we take $\overline{clk}$ from the previous stage of the clock tree. Thus, we obtain a new design, as shown in Fig. 4(b). If we consider delay of the inverter and NOR gate being roughly the same, the falling transitions of clk and $clk_{1\text{-}3}^*$ in the circuit will occur simultaneously. This design is synchronous in the sense that all flip flops are triggered in synchrony with the global clock.

We simulated the new design in Fig. 4(b) by SPICE 3f3 using 2 μ CMOS technology, which proved that the new design has an ideal logic operation. We also measure the power dissipation of two synchronous designs in Figs. 3(a) and 4(b). The power dissipation diagrams are shown in Fig. 5 and prove that the new design reduces the power dissipation by 22%.

V. Conclusion

The behavioral description of a clock is the basis to analyze its triggering action on flip flops. Based on it, two types of clock-gating were introduced to form a derived clock. We showed that the procedure for designing a derived clock could be systematized so as to isolate the triggered flip flop from the master clock in its idle cycles. The achieved power saving can be significant. However, the additional clock skew may lower the maximum operation frequency. Based on analyzing the timing relation in clock gating, we then presented a new technique for generating the derived clock, which is synchronous with the master clock. Circuit simulation proved the quality of the new derived clock and its capability to reduce power dissipation. More work is needed to develop a systematic design procedure and an algorithm for realizing the proposed design principles for clock gating in large sequential circuits. The engineering issues mentioned in [3] have thus been resolved

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for practical application, opening the path for widespread adoption of the clock-gating technique in low-power design of custom IC's.

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A Complete Operational Amplifier Noise Model: Analysis and Measurement of Correlation Coefficient

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***Abstract*—In contrast to the general operational amplifier (op amp) noise model widely used, we propose a more complete and applicable noise model, which considers the correlation between equivalent input voltage noise source e_n and current noise source i_n. Based on the super-position theorem and equivalent circuit noise theory, our formulae for the equivalent input noise spectrum density of an op amp noise are applied to both the inverting and noninverting input terminals. By measurement, we demonstrate that the new expressions are significantly more accurate. In addition, details of the measurement method for our noise model parameters are given. A commercial operational amplifier (Burr–Brown OPA37A) is measured by means of a low-frequency noise power spectrum measuring system and the measured results of its noise model parameters, including the spectral correlation coefficient (SCC), are finally given.**

***Index Terms*—Noise models, operational amplifiers, spectral correlation coefficient.**

I. INTRODUCTION

Recently, integrated operational amplifiers (op amps) have been used in more and more practical applications. With the continual improvement of their noise characteristics, they have been commonly found in the design of preamplifier circuits. For this reason, the calculation of the circuit noise of an op amp and its low-noise design are paid more attention than ever. At present, the noise models [1]–[3] of the overwhelming majority of op amps are illustrated as in Fig. 1(a) and (b).

The commonly accepted two-port noise model is in Fig. 1(a). The op amp is considered noiseless and the equivalent voltage noise source e_n and current noise source i_n are referred back to the input terminals. Fig. 1(b) is commonly adopted when the positive terminal is grounded. To simplify calculation, in some models only e_n is adopted and i_n is neglected [4], [5]. The advantage of these equivalent circuits is simplicity and convenience. However, in the area of small-signal detection, the requirements of noise specifications in the course of calculation and design of a low-noise circuit become higher. The shortcoming of Fig. 1(a) and (b) is obvious: the correlation between voltage noise source e_n and current noise source i_n is not considered, giving rise to inaccuracy.

At present, methods for measuring e_n and i_n [6], [7] use a small value of source resistance to measure an equivalent input voltage noise e_n and use a very large source resistance to measure an equivalent input current noise i_n. Because the correlation is not considered in this method, the measuring method is only an approximate solution. In fact, it can be calculated that the neglect of the correlation item can lead to, at most, a 40% measurement error [7]. Thus, it is commonly believed that the method can give only an approximate solution, and cannot give an accurate solution.

To solve this problem, a more complete op amp noise model is presented in this paper, based on Fig. 1(c), which considers the correlation between e_n and i_n for each input terminal and then the formula of equivalent input noise power spectrum density for the inverting and noninverting input terminals can be derived. With different source resistors, the noise model parameters of an op amp have been measured by means of a low-frequency noise measuring system and the noise model parameters, including the spectral correlation coefficient, are presented.

II. A COMPLETE NOISE MODEL AND ITS EQUIVALENT INPUT NOISE POWER SPECTRUM

In order to improve precision of the noise model, based on Fig. 1(a) and (b), we use one equivalent voltage noise source and one equivalent current noise source at each op amp input terminal in our model. Second, it should be pointed out that the correlation between e_n and i_n at each input terminal should be considered for completeness. Let $\gamma = \gamma_1 + j\gamma_2$ be the spectral correlation coefficient (SCC), given by $\gamma = S_{ei}(f)/\sqrt{S_e(f)S_i(f)}$, in which $S_e(f)$, $S_i(f)$ are the power spectral densities of the voltage noise e_n and current noise i_n, respectively, and $S_{ei}(f)$ is the cross-spectral density [8] between e_{n1} and i_{n1}. Also let $\gamma' = \gamma_1' + j\gamma_2'$ be the SCC between e_{n2} and i_{n2}, in which i_{n1} and i_{n2} are current noises at two input terminals of an op amp. Thus, it can be concluded that there is no correlation between them. Fig. 1(c) is a complete op amp noise model including eight parameters, i.e., e_{n1}, i_{n1}, $\gamma = \gamma_1 + j\gamma_2$, e_{n2}, i_{n2}, and $\gamma' = \gamma_1' + j\gamma_2'$, each of which varies with frequency. It is obvious that all these parameters cannot be calculated by use of internal noise sources of an op amp, for noise sources in an op amp are so many that it is very difficult to calculate them separately and accurately. However, they can be calculated by measuring equivalent input noise power spectrum with different source resistors. Now the relation between the eight parameters and equivalent input noise power spectrum can be derived as follows.

Let $Z_1 = R_1 + jX_1$, $Z_2 = R_2 + jX_2$ and $Z_f = R_f + jX_f$, $e_1^2 = 4\kappa TR_1$, $e_2^2 = 4\kappa TR_2$, $i_f^2 = 4\kappa T/R_f$, where e_1^2 and e_2^2 are the thermal noise spectrum of resistance R_1 and R_2, i_f^2 is the current noise spectrum of resistance R_f. According to Fig. 2(a), its equivalent noise circuit can be drawn as in Fig. 2(b).

According to the superposition theorem, the gain of each noise source can be calculated first and then the total output noise can be obtained by addition of each noise source power. Multiplication by the square of the noise bandwidth finally gives the output noise power

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